

MA XUCHU

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EDUCATION

Xi'an Jiaotong University, Xi'an, China

Master of Electrical Engineering | GPA: 3.59/4.0 (Rank 2/11)

Sep 2021 – Jul 2024

Southwest Jiaotong University, Chengdu, China

Bachelor of Electrical Engineering and Automation | GPA: 3.65/4.0 (Rank 8/98)

Sep 2017 – Jul 2021

RESEARCH EXPERIENCE

Deep Learning Methods for Transformer Condition Prediction and Fault Diagnosis Based on DGA

Jul 2022 – Jul 2024

Xi'an Jiaotong University | Main Researcher

- Proposed a hybrid CBAM-SIREN-LSTM model for dissolved gas concentration forecasting, achieving MAPE 2.86% and RMSE 1.87—outperforming LSTM, CNN-LSTM, and CBAM-LSTM baselines.
- Designed a CA-CNN fault diagnosis model with 50 engineered gas-ratio features; applied K-means SMOTE and 10-fold stratified validation, reaching 93.26% validation accuracy and 87.16% test accuracy.

Maximum Allowable Length of High-voltage Cable Installation

Jul 2021 – Dec 2022

Xi'an Jiaotong University | Main Researcher

- Modeled electrical, transportation, and construction constraints for HV cables.
- Determined maximum allowable laying length: 2810m (electrical constraint) and 1700m (case study).

Research on Parallel Control Technology of Metro Auxiliary Inverter

Jul 2020 – Jun 2021

Southwest Jiaotong University | Undergraduate Thesis Project

- Designed SVPWM with dual-loop voltage-current control, incorporating filter and PI controller.
- Verified feasibility of dual-loop SVPWM control and resistive droop control via MATLAB/Simulink simulations.

PUBLICATIONS

Contributed to the paper “Analysis of the optimal parameters for serpentine laying of high-voltage cables considering the influence of various factors,” presented at the *3rd International Conference on Electrical, Electronic Information and Communication Engineering (EEICE 2022)*, Guilin, China.

PROFESSIONAL EXPERIENCE

Xi'an Power Supply Company, State Grid Shaanxi Electric Power Co., Ltd.

Aug 2024 – Present

Engineer, Transmission Operation and Maintenance Center

Xi'an, China

- Worked on operation and maintenance of high-voltage transmission lines, gaining practical experience with power grid infrastructure, field inspection workflows, and equipment condition assessment.
- Participated in UAV-based autonomous inspection tasks, including transmission-line point cloud data acquisition and UAV flight route planning.

AWARDS AND HONORS

Special Scholarship for Academic Excellence (2022, 2023)

Excellent Postgraduate Cadre (2023)

Outstanding Graduate, Southwest Jiaotong University (2021)

First Prize, CSEE Cup National University Student Mathematical Modeling Competition (2018)

SKILLS

Programming: Python (PyTorch, NumPy, Pandas, Matplotlib, PySide6), Matlab, C++

Language: English (TOEFL iBT: 91 [R24, L27, S17, W23])